

AB POWER ENGINEERING

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AGRICULTURAL TRACTOR

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1. The ratio of the drawbar power and the wheel axle power.

- a. Lugging efficiency
- ☒ b. Traction efficiency
- c. Mechanical efficiency
- d. None of the above

2. The sum of all resistance acting on the tractor.

- a. Coefficient of traction
- ☒ b. Running resistance
- c. Rolling resistance
- d. None of the above

3. Running resistance of a tractor includes ____.

- a. rolling resistance
- b. air and acceleration resistance
- c. slope resistance
- ☒ d. All of the above

4. Weight transfer means ____.

- a. to move paddy from the field to the granary
- b. to transfer torrens title from the landlord to the tenant
- ☒ c. to shift implement's weight to the tractor's drive wheel in order to improve traction
- d. None of the above

5. The international testing center for tractor.

- a. International Rice Research Institute
- ☒ b. NEBRASKA
- c. International Farm Machinery Institute
- d. None of the above

6. A vehicle designed to pull, propel, and supply power to operate machinery used in agricultural operation.

- ☒ a. Agricultural tractor
- b. Automobile
- c. Payloader
- d. None of the above

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7. The stirring wheel of the tractor can be found ____.

- a. in front of the tractor
- b. at the rear of the tractor
- ☒ c. in front of the operator's seat
- d. None of the above

8. The substance that is added to the tractor for the purpose of changing traction and stability.

- ☒ a. Ballast
- b. Cage wheel
- c. Plow
- d. Hydraulic lift
- e. All of the above

9. The power drive found at the rear of the tractor that propels power implements and stationary machines.

- ☒ a. Power take-off drive
- b. Hydraulic system
- c. Differential gear
- d. None of the above

10. The standard rpm of a tractor's PTO drive.

- a. 600 rpm
- b. 200 rpm
- ☒ c. 540 rpm
- d. None of the above

11. A two-wheeled, hand-operated tractor used primarily as a substitute for draft animals.

- ☒ a. Power tiller
- b. Reaper
- c. Seeder
- d. None of the above

12. An attachment to a tractor's rear wheel to increase traction.

- a. Ballast
- ☒ b. Cagewheel
- c. Stirring wheel
- d. None of the above

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13. The part of the tractor that controls the travel direction of the machine.

- a. Rear wheel
- b. Front wheel
- c. Steering wheel
- d. None of the above

14. A tractor device that automatically raises or lowers the implement in operation to maintain a given draft.

- a. Automatic draft control
- b. Automatic position control
- c. 3-point linkage
- d. None of the above

15. A tractor operating at 3.5 mph develops a drawbar pull of 3000 pounds in pulling a trailer. Estimate its required drawbar horsepower.

- a. 22 hp
- b. 28 hp
- c. 32 hp
- d. None of the above

16. The specific fuel consumption of a tractor is 0.3 lbs per BHP-hour. If the heating value of the fuel is 19500 BTU/lb, what is the brake thermal efficiency of the tractor?

- a. 43.5%
- b. 57.2 %
- c. 61.3%
- d. None of the above

17. The gear reduction of a tractor located in its power train between the differential and the driving wheel that transmits power at low speed but large torque to the rear axles and wheels.

- a. Differential
- b. Universal joint
- c. Final drive
- d. None of the above

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18. The inherent characteristic of tractors to maintain or to produce increased drawbar pull even if the speed is reduced.

- a. Lugging ability
- b. Mechanical efficiency
- c. Coefficient of traction
- d. None of the above

19. The lower PTO speed of a standard four-wheel tractor is

- a. 540 rpm
- b. 450 rpm
- c. 640 rpm
- d. None of the above

20. The special arrangement of gears that permits driving member to rotate slower or faster than the other and permits operation of driving wheels at different speed for easy turn.

- a. Spur gear
- b. Planetary gear
- c. Differential
- d. None of the above

21. An under load tractor drive wheel rotates at 1200 revolution passing through a newly cultivated field. At no load, the tractor drive wheel rotates at 1125 revolution. What is the percent slippage of the tractor drive wheel?

- a. 6.25%
- b. 8.34%
- c. 9.97%
- d. None of the above

22. The standard power-take-off speed of a tractor.

- a. 100 rpm
- b. 540 rpm
- c. 1200 rpm
- d. None of the above

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23. The standard number of spline for a PTO shaft.

- a. 6 spline
- b. 21 spline
- c. 20 spline
- d. All of the above

24. The standard shaft diameter of a PTO shaft.

- a. 35 mm
- b. 40 mm
- c. 50 mm
- d. None of the above

25. If a farm tractor is rated at 80 horsepower, its metric equivalent is ____.

- a. 75 kw
- b. 60 kw
- c. 80 kw
- d. None of the above

26. The part of the tractor to which implements are hitched.

- a. PTO drive
- b. Final drive
- c. Drawbar
- d. None of the above

27. The coefficient of traction for wheel-type tractor is highest for ____.

- a. concrete road
- b. fine sand
- c. sandy loam soil
- d. All of the above

28. The part of a tractor that retards or stops one wheel during operation to be able to make short turn.

- a. Differential brake
- b. Hand brake
- c. Brake pedal
- d. All of the above

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29. The part of the tractor engine where pistons slide back and forth and is replaced or rebored when damaged.

- a. Cylinder block
- b. Cylinder liner
- c. Cylinder head
- d. All of the above

30. A trailing-type implement is hitched to the tractor at the tractor's ____.

- a. single hitch point system
- b. three-point hitch system
- c. power take off drive system
- d. None of the above

31. Wheel slip tests of a tractor showed that its wheel rotates by two-and-a-half revolutions at 20-meters distance. If the tractor wheel has 2-meter diameter, what is the percentage slip of the tractor wheel?

- a. 15.90 %
- b. 21.65 %
- c. 27.89%
- d. None of the above

32. The PTO drive of a tractor is used to power ____.

- a. moldboard plow
- b. chisel plow
- c. rotary plow
- d. All of the above

33. Four-wheel-drive tractors are ____.

- a. tractors that have four running wheels
- b. tractors that have four running wheels and the two rear wheels are driven by an engine
- c. tractors that have four running wheels and all wheels are driven by an engine
- d. None of the above

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34. The center-to-center distance between two front wheels or two rear wheels.

- a. ☒ Wheel tread
- b. ☐ Wheel base
- c. ☐ Wheel lugs
- d. ☐ None of the above

35. Hydraulic control levers on typical four- wheel tractors are located at the ____.

- a. ☐ front of the operator
- b. ☐ left side of the operator
- c. ☒ right side of the operator
- d. ☐ None of the above

36. Which of the following types of drawbar is used for tractors?

- a. ☐ Fixed drawbar
- b. ☐ Swinging drawbar
- c. ☐ Link drawbar
- d. ☒ All of the above

37. If the tractor wheel delivers a torque of 378.45 N-m when rotavating a paddy field, what is the force at its wheel if its diameter is 2 meters?

- a. ☐ 256.4 N
- b. ☒ 378.4 N
- c. ☐ 401.3 N
- d. ☐ None of the above

38. The PTO shaft profile for 35-mm nominal diameter shaft, 6 splines, 540- rpm rated speed.

- a. ☐ Involute spline
- b. ☒ Straight spline
- c. ☐ V-spline
- d. ☐ None of the above

39. The PTO shaft profile for 35-mm nominal diameter shaft, 21 splines, 1000- rpm rated speed.

- a. ☒ Involute spline
- b. ☐ Straight spline
- c. ☐ V-spline
- d. ☐ None of the above

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40. The automatic draft control in a modern farm tractor ____.

- a. ☐ automatically controls the amount of air passing through the radiator to regulate the operating temperature of the engine
- b. ☐ positions the implement at a constant height or depth relative to the tractor
- c. ☒ is the hydraulic implement hitch system which automatically raises or lowers a soil engaging implement to maintain its draft at a pre-selected value
- d. ☐ None of the above

41. The power measured at the flywheel or at the crankshaft of a four-wheel tractor.

- a. ☐ Rated horsepower
- b. ☐ Indicated horsepower
- c. ☒ Engine horsepower
- d. ☐ All of the above

42. A tractor weighs 2.5 tons and runs in a deep mud soil rotavating a paddy field. Estimate the rolling resistance of the tractor wheel during operation?

- a. ☐ 467 kgf
- b. ☒ 625 kgf
- c. ☐ 764 kgf
- d. ☐ None of the above

43. The distance between the supporting surface and the lowest point of the tractor.

- a. ☐ Tractor height
- b. ☒ Tractor ground clearance
- c. ☐ Tractor overall height
- d. ☐ None of the above

44. The speed, in revolution per minute, specified by the manufacturer.

- a. ☐ Rated horsepower
- b. ☐ Engine speed
- c. ☒ Rated engine speed
- d. ☐ All of the above

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45. The weight of the tractor includes its total weight, the different tools, the fuel tank filled to ____ capacity, and with the normal amount of cooling water and lubricating oil when the tractor is at work.

- a. 60%
- b. 80%
- c. 100%
- d. None of the above

46. The center of gravity of a standard two-wheel-drive tractor is located at ____.

- a. its mid center
- b. 1/3 of the wheel base in front of its rear wheel axle
- c. near its front axle wheel
- d. None of the above

47. A tractor pulls a 2.0meter-wide heavy-duty offset disk harrow at 10 kph operating speed. What is the theoretical field capacity of the machine?

- a. 2.0 ha/hr
- b. 2.4 ha/hr
- c. 3.1 ha/hr
- d. None of the above

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48. The center of gravity of a four-wheel drive tractor is located at ____.

- a. its mid center
- b. 1/3 of the wheel base in front of its rear wheel axle
- c. near its front axle wheel
- d. None of the above

49. John Deere tractor is made from ____.

- a. USA
- b. Japan
- c. Finland
- d. None of the above

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50. Massey Ferguson tractor is made from ____.

- a. Finland
- b. United Kingdom
- c. USA
- d. None of the above

51. Ford tractor is made from ____.

- a. United Kingdom
- b. USA
- c. France
- d. None of the above

52. Goldstar tractor is made from ____.

- a. Japan
- b. Korea
- c. China
- d. None of the above

53. Valtra Velmet tractor is made from ____.

- a. Finland
- b. Brazil
- c. France
- d. None of the above

54. Maxion tractor is made from ____.

- a. Finland
- b. Brazil
- c. France
- d. None of the above

55. If the specific fuel consumption of the tractor engine is 350 g of diesel/kW-hr during plowing operation, how much diesel will the 300 hp tractor will approximately consume in 8 hours operation? Assume a specific gravity of diesel of 0.76.

- a. 721.57 liters
- b. 769.24 liters
- c. 824.53 liters
- d. None of the above

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56. The indicated horsepower of the tractor engine is 300 hp. What is its expected power take off horsepower of the tractor?

- a. 241 hp
- b. 261 hp
- c. 345 hp
- d. None of the above

57. What is the drawbar power of the tractor in Item 56 above?

- a. 225 hp
- b. 298 hp
- c. 315 hp
- d. None of the above

58. The pivotal point of connection of hitch to the tractor or the center of the articulated connection between link and the implement.

- a. Link point
- b. Hitch pin
- c. Hitch point
- d. All of the above

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59. An external shaft of an agricultural tractor other than the PTO that provides rotational power to implements that are usually front- or side-mounted.

- a. Belt-drive shaft
- b. Secondary PTO
- c. Auxiliary PTO
- d. None of the above

60. The specific fuel consumption of a 250 hp tractor engine is 300 g of diesel/kw-hr. How many liters of fuel will the tractor consume?

- a. 70.82 liters per hour
- b. 79.65 liters per hour
- c. 86.52 liters per hour
- d. None of the above

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61. The effective drawbar pull of a tractor is reduced by ____ for every percent of grade.

- a. 0.1 %
- b. 1 %
- c. 10 %
- d. None of the above

62. A hitch system for a tractor suitable for large, heavy implement for trailing.

- a. Three-point hitch system
- b. Drawbar hitch system
- c. Semi-mounted hitch system
- d. All of the above

63. A device used to protect tractor operator from dust, noise, rain and heat.

- a. Operator's shield
- b. Chassis
- c. Cab
- d. None of the above

64. In item 60, Assume a specific gravity of fuel of 0.79 kg per liter and a cost of diesel of P47 per liter, how much fuel would it cost for 10 hours?

- a. P26,946.8
- b. P28,635.6
- c. P33,285.4
- d. None of the above

65. A type of tractor tire used to propel as well as to provide good floatation.

- a. Implement tire
- b. Traction tire
- c. Steering tire
- d. All of the above

66. The draft force of the tractor can be found at its ____.

- a. wheel
- b. hitch
- c. PTO drive
- d. None of the above

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67. The perpendicular distance between the upper hitch point and the common axis of the lower hitch point of the tractor.

- ☒ a. Mast height
- b. Hitch vertical height
- c. Stud height
- d. None of the above

68. A frame for protection of the operator of a compact-utility tractor to minimize the possibility of serious injury resulting from accidental upset.

- a. Roll bar
- b. Tractor shield frame
- ☒ c. Roll-over protection structure
- d. None of the above

69. A self-propelled machine with single axle designed primarily to pull and propel trailed or mounted agricultural implement and machinery.

- a. Hand tractor
- b. Pedestrian tractor
- c. Walking-type tractor
- ☒ d. All of the above

70. Fraction of power available that is delivered by the axle of the tractor to the implement through its hitch or drawbar.

- ☒ a. Tractive efficiency
- b. Pulling efficiency
- c. Drawbar efficiency
- d. None of the above

71. A neutral salt mixed with water and is added to the tire of the tractor to increase draft.

- ☒ a. Sodium chloride
- b. Calcium chloride
- c. Potassium iodide
- d. None of the above

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72. A walking-type agricultural tractor with 5.1- to 16-hp prime mover gasoline engine is classified as ____.

- a. Type 1
- ☒ b. Type 2
- c. Type 3
- d. None of the above

73. A walking-type agricultural tractor with a prime mover of up to 5-hp gasoline engine is classified as ____.

- ☒ a. Type 1
- b. Type 2
- c. Type 3
- d. None of the above

74. A tractor has the following engine specifications: number of cylinders, 6; cylinder bore, 0.15 m; piston stroke, 0.20 m; engine speed, 2200 rpm; mean effective pressure, 180 psi; and number of stroke, 4. What is its indicated horsepower?

- ☒ a. 646.7 hp
- b. 723.8 hp
- c. 791.3 hp
- d. None of the above

75. What is the brake horsepower of the above tractor engine (Item 74), assuming an 80% engine mechanical efficiency?

- a. 497.3 hp
- ☒ b. 517.4 hp
- c. 678.1 hp
- d. None of the above

76. What is the estimated PTO power of the tractor in Item 74 if its transmission efficiency is 90%?

- ☒ a. 465.7 hp
- b. 531.2 hp
- c. 591.4 hp
- d. None of the above

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77. What is the wheel axle power of the tractor if its transmission efficiency is 92%?

- a. 475.9 Hp
- b. 521.7 Hp
- c. 592.6 Hp
- d. None of the above

78. What is the drawbar power of a tractor if its transmission efficiency is 81%?

- a. 419.1 Hp
- b. 466.5 Hp
- c. 512.4 Hp
- d. None of the above

79. Considered as hazards and risks when using a tractor.

- a. Tractor rollover
- b. Tractor runover
- c. Tractor power takeoffs
- d. All of the above

80. The major expression of the power ratings of a tractor.

- a. Brake
- b. Drawbar
- c. Power Take Off
- d. All of the above
- e. None of the above

81. The maximum power the tractor engine can develop.

- a. Brake power
- b. Drawbar power
- c. Power take off power
- d. None of the above

82. The recommended level of wheel slip on a tilled soil for a four-wheel-drive tractor.

- a. 10%
- b. 11%
- c. 12%
- d. None of the above

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83. The coefficient of rolling resistance of a wheel-type tractor on a dry hard ground.

- a. 0.07
- b. 0.11
- c. 0.25
- d. None of the above

84. The coefficient of rolling resistance of a wheel-type tractor on a deep mud.

- a. 0.25
- b. 0.35
- c. 0.45
- d. None of the above

85. The recommended level of wheel slip on a tilled soil for a two-wheel-drive tractor.

- a. 10%
- b. 11%
- c. 12%
- d. None of the above

86. Indicators in determining the emission limit for a tractor.

- a. CO, HC, NOx, PM
- b. CO₂, HC, SOx, PM
- c. CO₂, HC, NOx, SOx, PM
- d. None of the above
- e. All of the above

87. The measure of the pulling power of the tractor engine by way of tires, wheels or tracts.

- a. Brake power
- b. Drawbar power
- c. Power take off power
- d. None of the above

88. Tractor implement control that automatically changes the position of the implement when there is an increase in the resistance in pulling and implement.

- a. Hydraulic draft control lever
- b. Hydraulic position control lever
- c. Remote control valve lever
- d. All of the above

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89. It is a manually operated control to change the implement position when there is an increase in implements draft.

- a. Hydraulic draft control
- b. Position draft control
- c. Mechanical control
- d. None of the above

90. If a four wheel tractor is equipped with rear and front wheel having lugs, it is a _____ tractor.

- a. rear-wheel
- b. front-wheel drive
- c. rear and front wheel drive
- d. All of the above

91. Tractor wheel diameter is determined based on the _____.

- a. rim diameter
- b. rubber diameter
- c. average of the the rim and rubber diameter
- d. None of the above

92. Tractor wheel width is determined based on the _____.

- a. rim width
- b. rubber width
- c. Average of rim and rubber width
- d. All of the above

93. When the tractor air cleaner is in horizontal position at the engine compartment, the air filter is a _____.

- a. dry-type filter
- b. wet-type filter
- c. no type filter
- d. None of the above

94. When the tractor air cleaner is in vertical position at the engine compartment, the air filter is a _____.

- a. dry-type filter
- b. wet-type filter
- c. no type filter
- d. None of the above

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95. Tractor ladders are commonly located at the _____ side of the driver seat.

- a. left
- b. right
- c. Rear
- d. All of the above

96. A tractor engine equipped with blower and turbine at the exhaust manifold of the engine cylinder.

- a. Naturally aspirated
- b. Super turbo
- c. Super turbo intercooled
- d. All of the above

97. A tractor engine equipped with blower and turbine at the exhaust manifold of the engine cylinder with a separate radiator attachment.

- a. Naturally aspirated
- b. Super turbo
- c. Super turbo intercooled
- d. All of the above

98. A tractor engine without blower, turbine, and separate radiator attachment at the exhaust manifold of the engine.

- a. Naturally aspirated
- b. Super turbo
- c. Super turbo intercooled
- d. All of the above

99. A counter weight provided in front of the the tractor.

- a. Front ballast
- b. Wheel ballast
- c. Implement weight
- d. None of the above

100. A weight provided at the rear wheel of the tractor to increase traction.

- a. Front ballast
- b. Wheel ballast
- c. Implement weight
- d. None of the ab

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101. Transmission type commonly used in four wheel tractor.

- a. Synchronesh
- b. Synchro shuttle
- c. Differential gear
- d. None of the above

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